Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/16/2005
PI Name:	Buchanan, Thomas Ph.D.		
Project Title:	Soot Formation in Purely-Curved Premixed Flames and Laminar Flame Speeds of Soot-Forming Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous flames - premixed		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19716-3140	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	12/01/2000	**End Date:**	11/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sung, Chih-Jen (Case Western Reserve University)		
Grant/Contract No.:	NCC3-843		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This joint program aims to gain fundamental understanding of the structure and dynamics of sooting premixed flames in simple, well-defined, one-dimensional, burner-supported, cylindrical and spherical flow fields. The proposed program consists of the following four major components: (1) experimental determination of the fundamental sooting limits in the absence of conductive heat loss to the burner; (2) experimental determination of the fundamental laminar flame speed of soot-forming flames; (3) experimental mapping of sooting structure; and (4) detailed modeling of sooting flame propagation and soot formation. The work in components 1 and 2 will demonstrate a novel experimental approach under microgravity and generate fundamental flame and soot data that are suitable for rigorous comparison with detailed numerical simulation. This experimental approach eliminates the ambiguities associated with the boundary conditions in previous soot laminar premixed flame experiments and simulation. Combined with components 3 and 4 of the proposed work, we shall develop a soot research platform which is unique and complementary to the existing soot research tools and facilities. To study rich, freely-propagating, sooting premix flames by microgravity experiments and numerical		

	simulation. This experiment is the first time that sooting flames are studied without the added complications of flame stretch or heat loss to the burner. The results of the study will yield fundemental insight into minimizing particulate pollutant emission from combustion.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Flame soot is a carcinogen and also contributes to global warming. Current work aims to further our anderstanding of sooting premixed flames, and improve current sooting models. This will allow better predictions of soot formation in practical combustion systems.
Task Progress:	Through this joint research program, we have gained fundamental understanding of the structure and dynamics of sooting premixed flames in simple, well-defined, one-dimensional, burner-supported, cylindrical and spherical flow fields. Experimentally, the fundamental sooting limits with/without metal additives and the laminar flame speeds of sooting flames have been determined. In addition, experimental mapping of sooting structures was conducted and detailed modeling of sooting flame propagation was made. The work conducted during the course of our study demonstrate a novel experimental approach under microgravity and generate fundamental flame and soot data that are suitable for rigorous comparison with detailed numerical simulation. This experimental approach eliminates the ambiguities associated with the boundary conditions in previous soot laminar premixed flame experiments and simulation.
Bibliography Type:	Description: (Last Updated: 04/29/2010)
Articles in Peer-reviewed Journals	Ibarreta AF, Sung CJ. "Flame temperature and location measurements of sooting remixed Bunsen flames by rainbow schlieren deflectometry." Appl Opt. 2005 Jun 10;44(17):3565-75. PubMed PMID: 16007857 , Jun-2005
Articles in Peer-reviewed Journals	Ibarreta AF, Sung CJ, Hirasawa T, Wang, H. "Burning velocity measurements of microgravity spherical sooting premixed flames using rainbow Schlieren deflectometry." Combustion and Flame 2005 Jan;140(1-2):93-102. http://dx.doi.org/10.1016/j.combustflame.2004.10.007 , Jan-2005
Articles in Peer-reviewed Journals	Zha0 B, Yang Z, Li Z, Johnston MV, Wang H. "Particle size distribution function of incipient soot in laminar premixed ethylene flames: effect of flame temperature." Proceedings of the Combustion Institute, 2005 Jan;30(1):1441-8. http://dx.doi.org/10.1016/j.proci.2004.08.104 , Jan-2005
Articles in Peer-reviewed Journals	Hirasawa T, Sung C-J, Yang Z, Joshi A, Wang H. "Effect of ferrocene addition on sooting limits in laminar premixed ethylene–oxygen–argon flames." Combustion and Flame 2004 Dec;139(4):288-99. http://dx.doi.org/10.1016/j.combustflame.2004.09.002 , Dec-2004
Presentation	Ibarreta, A.F., Sung, C.J., Hirasawa, T., and Wang, H "Burning Velocity Measurements of Microgravity Spherical Sooting Premixed Flames Using Rainbow Schlieren Deflectometry" Poster presented at the 30th International Symp. On Combustion, Chicago Jul-2004